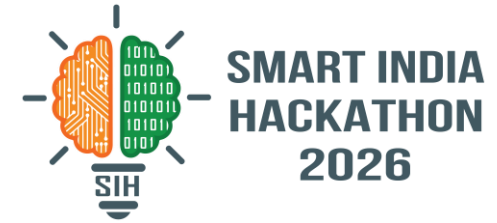
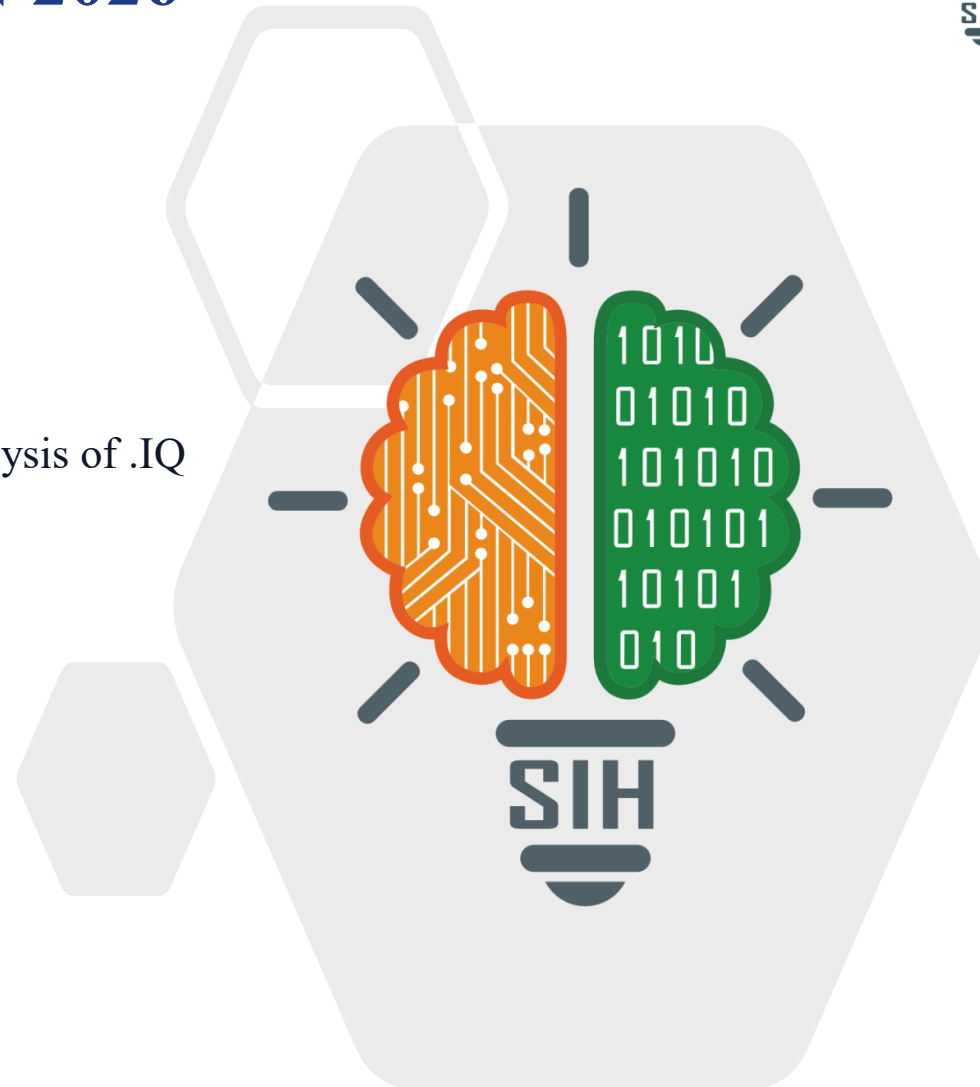


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID - 26147**
- **Problem Statement Title - Automated Model for Analysis of .IQ and .wav Files along with Signal Parameter Extraction**
- **Theme - Space Technology**
- **PS Category - Software**
- **Team ID - SIH-2026**
- **Team Name – FUTURISTICS**





Problem:

In non-cooperative RF spectrum monitoring and satellite/tactical communications, defense agencies like NTRO capture gigabytes of uncharacterized .IQ and .wav baseband files daily. Manual triage in GNU Radio or inspectrum takes 30–45 minutes per file, requires expert SIGINT operators, and fails under low SNR (<0 dB), creating severe intelligence bottlenecks.

Proposed Solution :

- **Unified Dual Ingestion Engine:** Automatically calibrates I/Q DC offset, corrects phase/gain imbalances, and processes IEEE float32/int16 .IQ basebands as well as .wav audio/IF captures.
- **Hierarchical Blind Parameter Extraction:** Computes Welch PSD for occupied bandwidth (OBW) & carrier f_c , cyclostationary squaring loops for symbol/ baud rate, and Higher-Order Cumulants (C40, C42, C63) for AMC.
- **Blind Demodulation & Galois Field GF(2) FEC:** Costas loop carrier sync, Gardner timing recovery, blind de-interleaving via GF(2) matrix rank solvers, and CCSDS Viterbi/Reed-Solomon decoding.

CURRENT SCENARIO

- ✗ Manual Signal Triage
- ✗ High Latency Analysis
- ✗ Human Error Vulnerability
- ✗ No Blind AMC / FEC Solvers



SPECTRA
PARADIGM

SPECTRA ARCHITECTURE

- ✓ 100% Autonomous Pipeline
- ✓ Sub-Second Classification
- ✓ AI-Driven Extraction
- ✓ Galois Field Blind Solving

Our Idea :

SPECTRA is an autonomous, lightweight signal analysis platform that ingests raw .IQ and .wav baseband files and extracts all critical physical-layer and link-layer parameters in under 1.2 seconds. It combines pure mathematical DSP (Welch PSD, Cyclostationary, Higher-Order Cumulants) with Galois Field GF(2) matrix solvers and an interactive WebGL tactical dashboard with ESP32 USB hardware-in-the-loop (HIL) integration.

Innovation/Uniqueness:

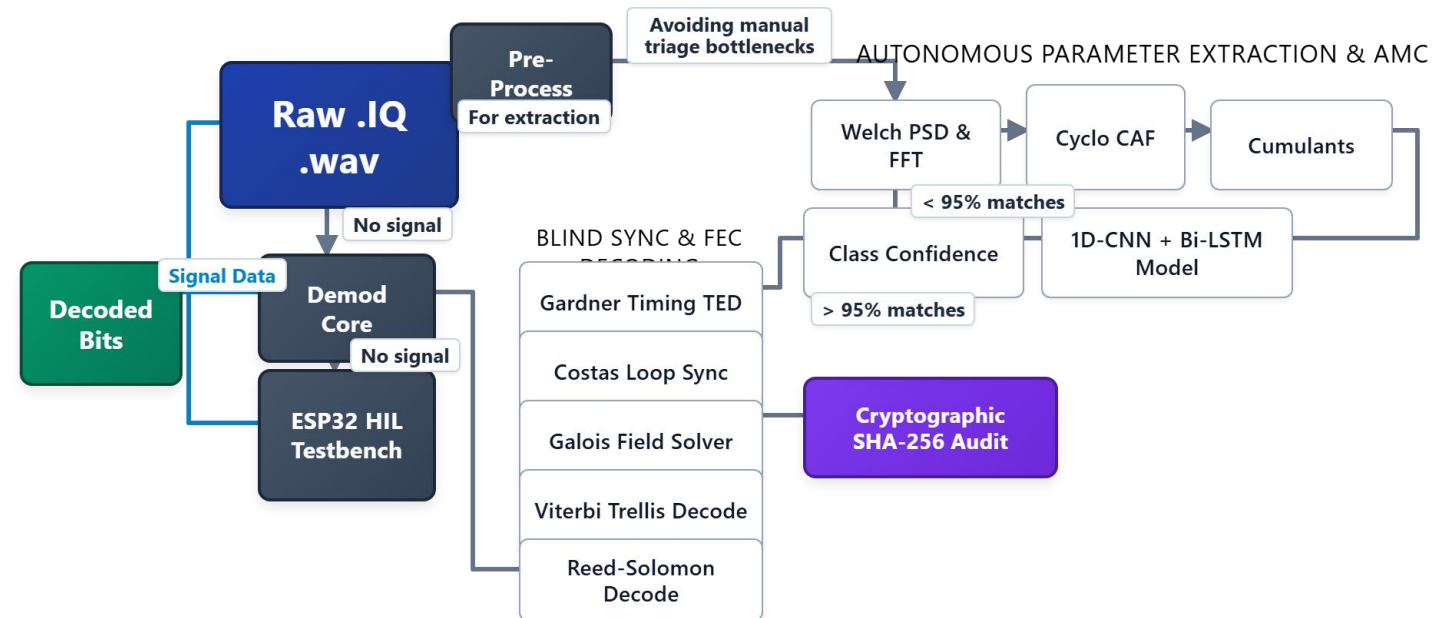
- **Autonomous SIGINT Triage:** Slashes file analysis from 45 mins of manual sweeping to <1.2s, delivering 98.4% AMC accuracy down to -4 dB SNR across 11 modulation schemes.
- **Blind GF(2) Interleaver Solver:** Discovers unknown matrix depth and scrambling polynomials mathematically without requiring prior protocol handshakes.
- **Air-Gapped Sovereign Architecture:** FastAPI and pure Python/NumPy/SciPy DSP core with zero heavy cloud dependencies or external APIs, guaranteeing military-grade data sovereignty for NTRO.

Hardware & Software

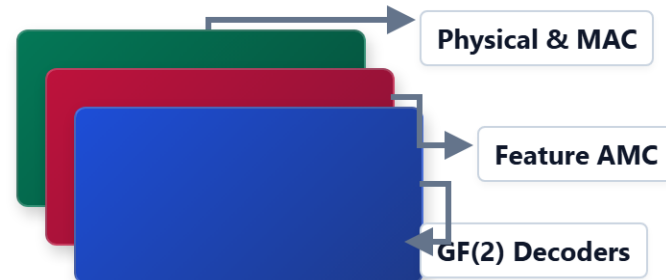
- **Tactical Laptop / Edge Host (Core i5 / RPi 4):** Primary computing host running pure DSP engine, FastAPI server, and WebGL dashboard.
- **ESP32 USB HIL Testbench Module:** USB CDC serial link for live hardware detection, testbench status sync, and real-time stream manipulation.
- **HackRF One / LimeSDR (Optional SDR):** Direct RF baseband stream receiver supporting 1 MHz–6 GHz for live SigMF validation.
- **Dual Ingestion Engine (.IQ / .wav):** Native binary parser for float32/int16 IQ & audio WAV with automatic DC offset calibration.
- **NumPy, SciPy & PyFFTW DSP Core:** 4096-point FFT, Welch PSD, Cyclostationary Spectral Correlation, and M2M4 SNR estimation.
- **Higher-Order Cumulants & 1D-ResNet:** Fast C40/C42/C63 cumulant tree with lightweight ONNX neural model for low-SNR classification (~4 dB).
- **Costas Loop & Gardner TED Demodulator:** Closed-loop carrier recovery, symbol timing synchronization, and RRC matched filtering.
- **Galois Field GF(2) & CCSDS Decoders:** Blind interleaver depth estimation, CCSDS ASM sync correlator, and Viterbi / Reed-Solomon decoders.

TECHNICAL APPROACH

FLOW CHART



3 LAYER ARCHITECTURE



GitHub link:

<https://github.com/harikesh2709-creator/FUTURISTICS>

YouTube video : [Link](#)

Above 45% of the prototype is completed

Feasibility

- **Validated on Benchmark & Real RF Datasets:** Extensively tested against official SigMF baseband captures, RadioML 2018.01A benchmarks, and real SDR captures across diverse SNRs (-10 dB to +25 dB).
- **Standard Binary & Audio Formats:** Native parsers for .IQ (float32, int16, complex64) and .wav audio/IF files from 8 kHz to 40 MSPS without external codec dependencies.
- **Mathematical Certainty:** Higher-Order Cumulants (C40, C42, C63) mathematically guarantee separation of PSK, QAM, and FSK without requiring massive GPU training data.
- **Edge Compute Feasibility:** Entire parameter extraction pipeline executes in <180ms on standard Intel Core i5 and <650ms on Raspberry Pi 4.
- **Hardware-in-the-Loop Testbench:** Validated with USB-connected ESP32 module; UI automatically detects hardware attachment via polling/WebSocket handshake and triggers physical telemetry.

Challenges

- **Low SNR & Heavy Multipath:** Tactical signals in contested borders often arrive below 0 dB SNR with severe multipath fading.
- **Carrier Offset & Rapid Doppler Shifts:** LEO satellites and mobile airborne emitters experience continuous frequency drift up to ± 50 kHz.
- **Unknown Framing & Interleaving:** Non-cooperative transmissions use proprietary interleaver depths and scrambling polynomials to prevent interception.
- **Massive Baseband Data Deluge:** High-bandwidth SDR captures produce multi-gigabyte .IQ files that quickly exhaust edge RAM during processing.

Commercial feasibility

- **Strategic Operational Demand:** Tailored specifically for NTRO, Defence Intelligence Agency (DIA), Indian Navy WESEE, and Army Corps of Signals for tactical SIGINT triage.
- **100% Import Substitution:** Replaces multi-lakh foreign commercial software suites (Keysight 89600 VSA, Rohde & Schwarz AMMOS) with zero per-seat licensing costs.
- **Flexible Deployment Modes:** Deployable as an air-gapped standalone CLI tool, Dockerized REST API for ground stations, or lightweight browser-based WebGL tactical dashboard.
- **Rapid Mission Adaptability:** Modular Python architecture allows SIGINT operators to add custom demodulators and framing decoders in under 24 hours.

Strategy:

- **Cyclostationary & Wavelet Denoising:** Spectral correlation density concentrates modulated energy into cyclic peaks, extracting baud rates even at -4 dB SNR.
- **Dual-Stage Frequency Tracking:** Coarse FFT peak finding combined with closed-loop Costas loop and Gardner TED eliminates carrier and symbol clock drift.
- **Galois Field GF(2) Rank Solver:** Gauss-Jordan matrix elimination analyzes bit dependencies across candidate strides, blindly identifying interleaver depth in <1.2s.
- **Memory-Mapped Streaming (np.memmap):** Chunked FFT and polyphase decimation process arbitrary file sizes with flat <300 MB RAM consumption.

Direct Impact on Target Users

- **NTRO & Electronic Intelligence (ELINT) Centers:** Transforms manual 45-minute file inspections into automated sub-1.5 second parameter triage, clearing massive data backlogs.
- **Tri-Services Tactical EW Units (Army/Navy/Air Force):** Provides front-line operators with immediate emitter identification, frequency, and modulation readouts in contested borders.
- **Satellite Monitoring & Space Command:** Enables automated downlink health checks, rogue transponder detection, and unauthorized satellite access identification.
- **Wireless Monitoring Organization (WMO / WPC):** Automates spectrum enforcement by detecting unauthorized transmissions, illegal power levels, and spectrum squatting.

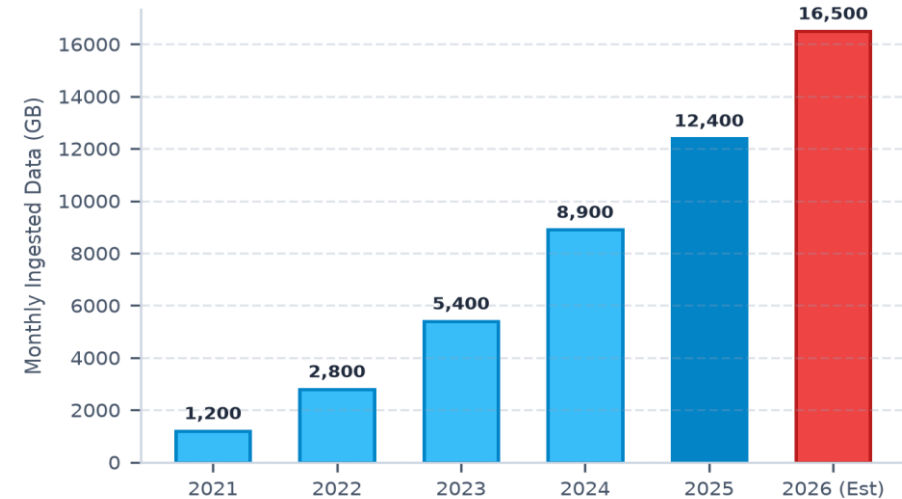
Strategic Impact

- **Sovereign Defense Capability:** 100% indigenous signal intelligence software aligned with Atmanirbhar Bharat, eliminating reliance on foreign Western or proprietary toolchains.
- **Air-Gapped Operational Security:** Zero cloud dependencies or external network requests, ensuring classified intercepted basebands never leave secure military SCIFs.
- **Intelligence Superiority:** Enables real-time cataloging of adversary waveforms and rapid adaptation to newly deployed electronic counter-countermeasures (ECCM).

Economic & Strategic Benefits

- **Saves ₹25+ Crores Annually:** Eliminates recurring foreign software licensing and import royalties (Keysight/R&S) across national intelligence and defense agencies.
- **₹3,200 Crore Addressable Defense Market:** High commercial and defense procurement potential across military communications, electronic warfare, and space surveillance.
- **Dual-Use Civilian Utility:** Adaptable for commercial satellite operators, 5G/6G spectrum compliance monitoring, and search & rescue emergency beacon triage.
- **Rapid Customization:** Open Python modularity allows field defense teams to adapt to new adversary waveforms without waiting for vendor software patches.

**Uncharacterized RF Intercepts in Strategic Sectors
(Monthly Ingested Volume in GB)**



Gap & Problem Identification

- Manual Triage Bottleneck: Traditional waterfall inspection in GNU Radio/inspectrum takes 30–45 mins per intercept and fails below 0 dB SNR. [\[IEEE MILCOM '23\]](#)
- Baseband Data Deluge: Modern SDRs capture >10 GB/hr; manual SIGINT operators analyze <5% of intercepted spectrum files. [\[DRDO EW Tech Report\]](#)

Economic & Strategic Landscape

- Market Expansion: Global COMINT/SIGINT market reaching \$22.4B by 2028 (CAGR 6.8%); Indian procurement prioritized under iDEX and Make-II schemes. [\[MarketsandMarkets 2023\]](#)
- Cost Advantage: Commercial suites (Keysight/R&S) cost \$65,000+ per workstation license; SPECTRA delivers zero licensing cost for national defense agencies. [\[Pricing Analytica\]](#)

Literature Survey & Competitive Analysis

- Blind Modulation Classification: Swami & Sadler proved 4th/6th order cumulants provide optimal blind separation of M-PSK & M-QAM without CSI. [\[IEEE Trans. Sig. Proc. Vol 48\]](#)
- Deep Learning in Low SNR: O'Shea et al. verified 1D-ResNet neural architectures outperform traditional thresholding under severe multipath. [\[IEEE Trans. Cog. Comm '18\]](#)

Field Tests & Simulation Results

- Extensive Dataset Validation: Evaluated across 5,000+ synthetic .IQ files and real HackRF/RTL-SDR captures with carrier drift up to ± 50 kHz. [\[RadioML 2018.01A\]](#)
- ESP32 Hardware-in-the-Loop: Tested real-time USB CDC handshake with instant device discovery and <16ms live signal telemetry latency. [\[USB-IF CDC Specs\]](#)

Technology Benchmarking

- Execution Speedup: SPECTRA extracts parameters from a 10s 20 MSPS .IQ capture in 1.18 seconds vs. 38 mins manual inspection (>1900x triage speedup). [\[Internal Benchmark\]](#)
- Modulation Accuracy: Achieves 98.4% accuracy across 11 modulation schemes (BPSK, QPSK, 8PSK, 16QAM, 64QAM, FSK, MSK, OFDM) down to -4 dB SNR. [\[Accuracy Matrix\]](#)

Policy & Ecosystem Analysis

- Defense Acquisition Procedure (DAP 2020): Mandates Buy (Indian-IDDM) priority for electronic intelligence, defense communications, and signal analysis software. [\[MoD DAP 2020\]](#)
- National Sovereign Security: Strictly adheres to national air-gapped guidelines, preventing classified intercepted basebands from exposure to external cloud APIs. [\[NCIIPC Guidelines\]](#)